

API Endpoint Plan

Authentication Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Register a new user account with role selection (Organiser or Participant)	None (Public)	{ "email": "string", "password": "string", "confirmPassword": "string", "firstName": "string", "lastName": "string", "phone": "string", "dateOfBirth": "date", "role": "string" }	201 Created - { "userId": "int", "email": "string", "firstName": "string", "lastName": "string", "role": "string", "token": "string" } 400 Bad Request - validation errors 409 Conflict - email already exists
POST	/api/auth/login	Authenticate user and establish session	None (Public)	{ "email": "string", "password": "string" }	200 OK - { "userId": "int", "email": "string", "firstName": "string", "lastName": "string", "role": "string", "token": "string" } 401 Unauthorized - invalid credentials 403 Forbidden

					- account inactive
POST	/api/auth/logout	Log out current user and clear session	Any	None	200 OK - { "message": "Logged out successfully" }
GET	/api/auth/me	Get current authenticated user's profile	Any	None	200 OK - user profile object 401 Unauthorized - not authenticated
PUT	/api/auth/me	Update current user's profile	Any	{ "firstName": "string", "lastName": "string", "phone": "string", "dateOfBirth": "date" }	200 OK - updated user profile 400 Bad Request - validation errors 401 Unauthorized - not authenticated
PUT	/api/auth/password	Change user's password	Any	{ "currentPassword": "string", "newPassword": "string", "confirmNewPassword": "string" }	200 OK - { "message": "Password updated successfully" } 400 Bad Request - invalid current password 401 Unauthorized - not authenticated
POST	/api/auth/forgot-password	Request password reset link	None (Public)	{ "email": "string" }	200 OK - { "message": "Password reset link sent if email exists" } 400 Bad

					Request - invalid email
POST	/api/auth/reset-password	Reset password using token	None (Public)	{ "token": "string", "newPassword": "string", "confirmPassword": "string" }	200 OK - { "message": "Password reset successfully" } 400 Bad Request - invalid or expired token
POST	/api/auth/refresh-token	Refresh authentication token	Any	{ "refreshToken": "string" }	200 OK - new token pair 401 Unauthorized - invalid refresh token

User Profile Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users	Get paginated list of all users with filters	Admin/Organiser	None (Query params: page, size, role, search)	200 OK - paginated list of user objects 403 Forbidden - insufficient permissions
GET	/api/users/{id}	Get a specific user's profile by ID	Any (own profile) / Admin	None	200 OK - user object with role 403 Forbidden - cannot view other users (unless Admin) 404 Not Found - user doesn't exist
GET	/api/users/{id}/events	Get all events created by an organiser	Any	None (Query params: status,	200 OK - array of event objects 404 Not Found

				fromDate, toDate)	- user doesn't exist
GET	/api/users/{id}/enrolments	Get all event enrolments for a participant	Participant (own) / Organiser (own events)	None (Query params: status, fromDate, toDate)	200 OK - array of enrolment objects with event details 403 Forbidden - cannot view other users' enrolments 404 Not Found - user doesn't exist
GET	/api/users/{id}/results	Get all race results for a participant	Participant (own) / Any (public)	None (Query params: fromDate, toDate, eventId)	200 OK - array of result objects with event and category details 404 Not Found - user doesn't exist
POST	/api/users/{id}/profile-picture	Upload profile picture	Participant (own)	multipart/form-data - image file	200 OK - { "profilePictureUrl": "string" } 400 Bad Request - invalid file format 403 Forbidden - not own profile 404 Not Found - user doesn't exist
DELETE	/api/users/{id}/profile-picture	Remove profile picture	Participant (own)	None	204 No Content 403 Forbidden - not own profile 404 Not Found - user doesn't exist

Event Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Get all events with optional filters	None (Public)	None (Query params: status, eventType, location, dateFrom, dateTo, search, page, size)	200 OK - paginated array of event objects with organiser details 400 Bad Request - invalid query parameters
GET	/api/events/{id}	Get a specific event by ID with full details	None (Public)	None	200 OK - event object with categories, organiser, weather, and enrolment stats 404 Not Found - event doesn't exist
POST	/api/events	Create a new event	Organiser	{ "eventName": "string", "description": "string", "location": "string", "venue": "string", "eventDate": "datetime", "registrationDeadline": "datetime", "entryFee": "decimal", "maxParticipants": "integer", "eventType": "string", "distance": "string", "categories": [{"categoryId": "int", "fee": "decimal", "capacity": "int"}]}	201 Created - event object with categories 400 Bad Request - validation errors 401 Unauthorized - not authenticated

PUT	/api/events/{id}	Update an existing event	Organiser (owner),	{ "eventName": "string", "description": "string", "location": "string", "venue": "string", "eventDate": "datetime", "registrationDeadline": "datetime", "entryFee": "decimal", "maxParticipants": "integer", "status": "string" }	200 OK - updated event object 400 Bad Request - validation errors 403 Forbidden - not event owner 404 Not Found - event doesn't exist
DELETE	/api/events/{id}	Delete an event (soft delete - set IsDeleted flag)	Organiser (owner)	None	204 No Content 403 Forbidden - not event owner 404 Not Found - event doesn't exist
PATCH	/api/events/{id}/status	Update event status	Organiser (owner)	{ "status": "string" }	200 OK - updated event object 400 Bad Request - invalid status transition 403 Forbidden - not event owner 404 Not Found - event doesn't exist
POST	/api/events/{id}/banner	Upload event banner image	Organiser (owner)	multipart/form-data - image file	200 OK - { "bannerImageUrl": "string" } 400 Bad Request - invalid file format 403 Forbidden - not event owner 404 Not Found - event doesn't exist

DELETE	/api/events/{id}/banner	Remove event banner image	Organiser (owner)	None	204 No Content 403 Forbidden - not event owner 404 Not Found - event doesn't exist
GET	/api/events/{id}/stats	Get event statistics (enrolments, categories, etc.)	Any	None	200 OK - { "totalEnrolments": "int", "categoriesCount": "int", "availableSlots": "int", "percentageFull": "decimal", "enrolmentsByCategory": [] } 404 Not Found - event doesn't exist

Category Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/categories	Get all categories with optional filters	None (Public)	None (Query params: categoryType, distance, isActive)	200 OK - array of category objects
GET	/api/categories/{id}	Get a specific category by ID	None (Public)	None	200 OK - category object 404 Not Found - category doesn't exist
POST	/api/categories	Create a new category	Admin/Organiser	{ "categoryName": "string", "categoryType":	201 Created - category

				"string", "distance": "string", "ageGroup": "string", "gender": "string", "standardFee": "decimal", "minAge": "int", "maxAge": "int" }	object 400 Bad Request - validation errors 403 Forbidden - insufficien t permissio ns
PUT	/api/categories/{ id}	Update an existing category	Admin/Organis er	{ "categoryNam e": "string", "categoryType": "string", "distance": "string", "ageGroup": "string", "gender": "string", "standardFee": "decimal", "minAge": "int", "maxAge": "int", "isActive": "boolean" }	200 OK - updated category object 400 Bad Request - validation errors 403 Forbidden - insufficien t permissio ns 404 Not Found - category doesn't exist
DELETE	/api/categories/{ id}	Delete a category	Admin	None	204 No Content 403 Forbidden - insufficien t permissio ns 404 Not Found - category

					doesn't exist
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Event Category Endpoints (Managing Event-Category Links)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Get all categories for a specific event	None (Public)	None	200 OK - array of category objects with fee and capacity 404 Not Found - event doesn't exist
POST	/api/events/{eventId}/categories	Add a category to an event	Organiser (owner)	{ "categoryId": "int", "fee": "decimal", "capacity": "int" }	201 Created - event category object 400 Bad Request - validation errors or category already linked 403 Forbidden - not

					event owner 404 Not Found - event or category doesn't exist
PUT	/api/events/{eventId}/categories/{categoryId}	Update event category details (fee, capacity)	Organiser (owner)	{ "fee": "decimal", "capacity": "int" }	200 OK - update event category object 400 Bad Request - validation errors 403 Forbidden - not event owner 404 Not Found - event or category mapping doesn't exist
DELETE	/api/events/{eventId}/categories/{categoryId}	Remove a category from an event	Organiser (owner)	None	204 No Content 403 Forbidden - not event owner 404 Not Found -

					event or category mapping doesn't exist
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Enrolment Endpoints (Participant Registrations)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/enrolments	Get all enrolments with optional filters	Organiser/Admin	None (Query params: eventId, participantId, status)	200 OK - array of enrolment objects with participant and event details 403 Forbidden - insufficient permissions
GET	/api/enrolments/{id}	Get a specific enrolment by ID	Participant (own) / Organiser (own events)	None	200 OK - enrolment object with full details 403 Forbidden - cannot view other users'

					enrolments 404 Not Found - enrolment doesn't exist
POST	/api/enrolments	Create a new enrolment (participant enters event)	Participant	{ "eventId": "int", "eventCategoryId": "int" }	201 Created - enrolment object with registration number 400 Bad Request - validation errors, event full, or already enrolled 403 Forbidden - registration deadline passed 404 Not Found - event or category doesn't exist
PUT	/api/enrolments/{id}	Update an existing enrolment	Participant (own) / Organiser (own events)	{ "status": "string", "paymentStatus": "string" }	200 OK - updated enrolment object 400 Bad Request

		(status, payment)			- validation errors 403 Forbidden - cannot modify other users' enrolments 404 Not Found - enrolment doesn't exist
DELETE	/api/enrolments/{id}	Cancel an enrolment	Participant (own) / Organiser (own events)	None	204 No Content 400 Bad Request - cannot cancel completed event 403 Forbidden - cannot cancel other users' enrolments 404 Not Found - enrolment doesn't exist
GET	/api/enrolments/event/{eventId}	Get all enrolments for a	Organiser (owner)	None (Query params:	200 OK - array of enrolme

		specific event		status, categoryId)	nt objects with participant details 403 Forbidden - not event owner 404 Not Found - event doesn't exist
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Result Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/results	Get all results with optional filters,	None (Public)	None (Query params: eventId, participantId, categoryId)	200 OK - array of result objects with participant and event details
GET	/api/results/{id}	Get a specific result by ID	None (Public)	None	200 OK - result object with full details 404 Not Found - result doesn't exist

POST	/api/results	Create a new result	Organiser (owner)	{ "enrolmentId": "int", "gunTime": "time", "chipTime": "time", "overallPosition": "int", "categoryPosition": "int", "pace": "float", "status": "string" }	201 Created - result object 400 Bad Request - validation errors 403 Forbidden - not event owner 404 Not Found - enrolment doesn't exist
PUT	/api/results/{id}	Update an existing result	Organiser (owner)	{ "gunTime": "time", "chipTime": "time", "overallPosition": "int", "categoryPosition": "int", "pace": "float", "status": "string" }	200 OK - updated result object 400 Bad Request - validation errors 403 Forbidden - not event owner 404 Not Found - result doesn't exist

DELETE	/api/results/{id}	Delete a result	Organiser (owner)	None	204 No Content 403 Forbidden - not event owner 404 Not Found - result doesn't exist
GET	/api/results/event/{eventId}/leaderboard	Get leaderboard for a specific event	None (Public)	None (Query params: categoryId, limit)	200 OK - array of ranked results with positions 404 Not Found - event doesn't exist
GET	/api/results/participant/{participantId}	Get all results for a specific participant	Participant (own)	None	200 OK - array of result objects with event and category details 404 Not Found - participant doesn't exist

